

```
!pip install faiss-cpu transformers torch pandas
```

```
Requirement already satisfied: faiss-cpu in /usr/local/lib/python3.12/dist-packages (1.13.2)
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: numpy<3.0,>=1.25.0 in /usr/local/lib/python3.12/dist-packages (from faiss-cpu) (2.0.2)
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Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.24.3)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.5.0)
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Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.0)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
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Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
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Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggi
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0-
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Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggi
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Requirement already satisfied: mdurl~0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->
```

```
from google.colab import files
```

```
uploaded = files.upload()
```

No file chosen

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving links.csv to links (1).csv

Saving movies.csv to movies (1).csv

Saving ratings.csv to ratings (1).csv

Saving README.txt to README (1).txt

```
import pandas as pd
```

```
file_name = list(uploaded.keys())[0]
```

```
df = pd.read_csv(file_name)
```

```
print("Dataset loaded successfully")
```

```
print(df.head())
```

```
Dataset loaded successfully
```

	movieId	imdbId	tmdbId
0	1	114709	862.0
1	2	113497	8844.0
2	3	113228	15602.0
3	4	114885	31357.0
4	5	113041	11862.0

```
TEXT_COLUMN = df.columns[0] # safest default for exam
```

```
item_texts = df[TEXT_COLUMN].astype(str).tolist()
```

```
print("Total items:", len(item_texts))
```

```
print("Sample item:", item_texts[0])
```

Total items: 9742
Sample item: 1

```
from transformers import AutoTokenizer, AutoModel
import torch
import numpy as np
import faiss

device = "cuda" if torch.cuda.is_available() else "cpu"

model_name = "sentence-transformers/all-MiniLM-L6-v2"

tokenizer = AutoTokenizer.from_pretrained(model_name)
encoder = AutoModel.from_pretrained(model_name).to(device)
encoder.eval()
```

Loading weights: 100% 103/103 [00:00<00:00, 290.05it/s, Materializing param=pooler.dense.weight]

BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2

Key	Status
embeddings.position_ids	UNEXPECTED

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

```
BertModel(
  (embeddings): BertEmbeddings(
    (word_embeddings): Embedding(30522, 384, padding_idx=0)
    (position_embeddings): Embedding(512, 384)
    (token_type_embeddings): Embedding(2, 384)
    (LayerNorm): LayerNorm((384,), eps=1e-12, elementwise_affine=True)
    (dropout): Dropout(p=0.1, inplace=False)
  )
  (encoder): BertEncoder(
    (layer): ModuleList(
      (0-5): 6 x BertLayer(
        (attention): BertAttention(
          (self): BertSelfAttention(
            (query): Linear(in_features=384, out_features=384, bias=True)
            (key): Linear(in_features=384, out_features=384, bias=True)
            (value): Linear(in_features=384, out_features=384, bias=True)
            (dropout): Dropout(p=0.1, inplace=False)
          )
          (output): BertSelfOutput(
            (dense): Linear(in_features=384, out_features=384, bias=True)
            (LayerNorm): LayerNorm((384,), eps=1e-12, elementwise_affine=True)
            (dropout): Dropout(p=0.1, inplace=False)
          )
        )
        (intermediate): BertIntermediate(
          (dense): Linear(in_features=384, out_features=1536, bias=True)
          (intermediate_act_fn): GELUActivation()
        )
        (output): BertOutput(
          (dense): Linear(in_features=1536, out_features=384, bias=True)
          (LayerNorm): LayerNorm((384,), eps=1e-12, elementwise_affine=True)
          (dropout): Dropout(p=0.1, inplace=False)
        )
      )
    )
  )
  (pooler): BertPooler(
    (dense): Linear(in_features=384, out_features=384, bias=True)
    (activation): Tanh()
  )
)
```

```
def encode_texts(texts, batch_size=16):
    all_embeddings = []

    with torch.no_grad():
        for i in range(0, len(texts), batch_size):

            batch = texts[i:i + batch_size]

            tokens = tokenizer(
                batch,
                padding=True,
                truncation=True,
                return_tensors="pt"
```

```

        ).to(device)

        outputs = encoder(**tokens)

        embeddings = outputs.last_hidden_state.mean(dim=1)
        embeddings = embeddings.cpu().numpy().astype("float32")

        all_embeddings.append(embeddings)

    return np.vstack(all_embeddings)

item_embeddings = encode_texts(item_texts)

faiss.normalize_L2(item_embeddings)

print("Embeddings shape:", item_embeddings.shape)

```

Embeddings shape: (9742, 384)

```

embedding_dim = item_embeddings.shape[1]

index = faiss.IndexFlatIP(embedding_dim)

index.add(item_embeddings)

print("FAISS index built with", index.ntotal, "items")

```

FAISS index built with 9742 items

```

def get_similar_items(item_id, k=5):

    query_vector = item_embeddings[item_id:item_id+1]

    scores, indices = index.search(query_vector, k)

    return scores[0], indices[0]

```

```

query_id = 0

scores, idxs = get_similar_items(query_id, k=5)

print("QUERY ITEM:")
print(item_texts[query_id])

print("\nSIMILAR ITEMS:")

for score, idx in zip(scores, idxs):
    print(f"-> {item_texts[idx]} (score={score:.3f})")

```

QUERY ITEM:
1

SIMILAR ITEMS:
-> 1 (score=1.000)
-> 2 (score=0.805)
-> 3 (score=0.791)
-> 4 (score=0.769)
-> 7 (score=0.658)

